# QUESTION BANK

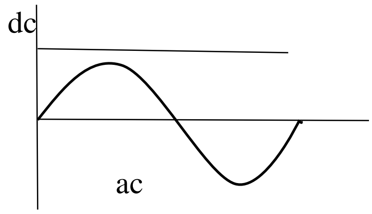
# CHAPTER: ALTERNATING CURRENT

TOPICS COVERED: Alternating currents, peak and rms value of alternating current/ voltage, reactance and impedance.

## SECTION A

## COMPETENCY BASED QUESTIONS

Q1.An alternating current is that whose magnitude changes continuously with time and direction reverses periodically why the current that flows with same magnitude in same direction is direct current,




(i)If an ammeter is used to measure alternating current in circuit, which value of current will it read

(a) Peak (b) rms (c) Instantaneous (d) Average

(ii) If T is the time period of given alternating current, then how much time it requires to reach its peak value starting from zero

(a)T (b) T/2 (c) T/4 (d) T/8

(iii) What is the average value of alternating current over one complete cycle

(a)0 (b) $I_o$ (c) $I_o/2$ (d) $I_o/4$

(iv)For a 220 V mains of alternating voltage peak value will be……………………

220 V b)220/ √2 V c) 220 √2 V d) 110 V

(v)How will you write an equation to represent alternating voltage with frequency 10 cycles per second and amplitude 100?

Q2. A DC ammeter cannot be used to measure an alternating current because

(a)DC ammeter will get damage
(b) average value of AC for complete cycle is zero
(c) alternating current cannot pass through DC ammeter
(d) alternating current changes its polarity

Q3. Why the use of ac voltage is preferred over dc voltage?

(a) Generation of ac is more economical than dc
(b) AC can be stepped up and stepped down
(c) It can be transmitted with lower energy loss
(d) All of the above

## Assertion & Reasoning

(A) Assertion A and reason R are both true and R is the correct explanation of A

(B) Assertion A and reason R are both true but R is not the correct explanation of A

(C) Assertion A is true but reason R is false

(D) Both assertion A and reason R are false

Q4. A: An alternating EMF leads alternating current by phase angle $\pi/2$ in a purely inductive circuit.

R: Inductive reactance decreases when frequency of source increases.

Q5. A: Alternating current is more dangerous than direct current.

R: Alternating current changes its polarity periodically.

Q6. A: A capacitor block dc but allow ac to pass through it.

R: Capacitive reactance is inversely proportional to frequency.

Q7. A: In an ac circuit having and inductance coil, the frequency of ac is increased, current decreases.

R: Current is inversely proportional to frequency.

## SA-I (2 MARKS EACH)

Q8.(i) Define root mean square value of an alternating current.

(ii) Write the relation between peak and root mean square value of alternating current.

Q9.The EMF of an AC source is given as: $\mathcal{E} = 100 \sin 314\, t$

Find peak value of alternating emf and frequency.

Q10.A light bulb is rated as 50 W for 220 V AC supply of 50 Hz, Calculate resistance of the bulb and rms current through the bulb.

Q11.Draw the graphs showing variation of inductive reactance and capacitive reactance with frequency of applied ac source.

Q12. The reactance of a capacitor at 50 Hz is 10 ohms, if frequency is doubled, then what will be the value of reactance?

## SA-II (3 MARKS EACH)

Q13.What is inductive reactance? If an ac voltage is applied across an inductance L, find an expression for the current I flowing in the circuit. Also show the phase relationship between current and voltage in a phasor diagram.

Q14.What is capacitive reactance? If an ac voltage is applied across a capacitance C, find an expression for the current I flowing in the circuit. Also show the phase relationship between current and voltage in a phasor diagram.

Q15.If the frequency of alternating current is tripled, how will it affect resistance R, inductive reactance $X_L$ and capacitive reactance $X_C$?

Q16.For a given ac circuit, differentiate between resistance, reactance and impedance.

## LA (5 MARKS)

Q17.(a) Differentiate between the term inductive reactance and capacitive reactance of an ac circuit.

(b) If an inductor and a resistor are connected in series in an ac circuit, what will be the mathematical expression for the impedance of this circuit. How will the impedance get affected when the frequency of applied signal is decreased and why?

# ANSWER KEY

## SECTION A

1.(i) b) rms
(ii) c) T/4
(iii) a) 0
(iv) c) 220 √2 V
(v) $\mathcal{E} = 100 \sin 20 \pi t$

2. (b) average value of AC for complete cycle is zero

3. (d) All of the above

4. (c) Assertion A is true but reason R is false

5. (b) Assertion A and reason R are both true but R is not the correct explanation of A

6. (a) Assertion A and reason R are both true and R is the correct explanation of A

7. (a) Assertion A and reason R are both true and R is the correct explanation of A

8. (i) root mean square is that value of current which produces the same heating effect in a given resistor as is produced by the given alternating current when passed for the same time. It is denoted by $I_{rms}$.

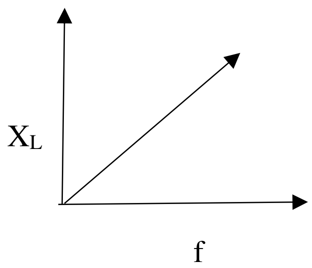
ii) $I_{rms} = I_{peak} / \sqrt{2} = 0.707\ I_{peak}$

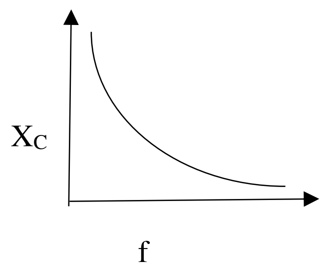
9. peak value of alternating emf = 100 V , for frequency: $2 \pi f = 314$, so f = 50 Hz

10. Resistance of bulb $R = V^2_{rms} / P = (220 \times 220)/ 50 = 968$ ohms

$I_{rms} = P / V_{rms} = 50/220 = 0.227$ A

11.




12. Capacitive reactance, $X_C = 1/ wC = 1/ (2 \pi f C) = 1/ 100 \pi C$

$X_C$, capacitive reactance when frequency is doubled $= 1/ (2 \pi (2f) C) = 1/ 200 \pi C = X_C/ 2$

13. Inductive reactance is the opposition offered by the inductor in flow of ac. It is denoted by $X_L$.

Let $\mathcal{E} = \mathcal{E}_o \sin wt$ be the emf of ac source connected with inductor in the circuit

Then an alternating current flows through the inductor, a back emf $-\ L\frac{dI}{dt}$ is set up which opposes the applied emf.

Net instantaneous emf = $\mathcal{E} - L\, dI/dt$

But this emf must be zero because there is no resistance in the circuit

So, $$\mathcal{E} - L\frac{dI}{dt} = 0$$

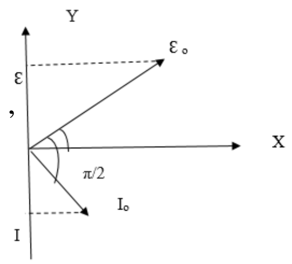
$$\mathcal{E} = L\frac{dI}{dt}$$

$$\mathcal{E}_o \sin w\, t = L\frac{dI}{dt}$$

$$dI = \frac{\mathcal{E}o}{L}\sin wtdt$$

Integrating, $$\int dI = \int \frac{\mathcal{E}o}{L}\sin wtdt$$

$$I = -\frac{\mathcal{E}o}{wL}\cos wt + \text{constant}$$




Over the time period T, cos wt = 0 and average value of I must be 0 so ,

integration constant = 0 hence, $I = -\frac{\mathcal{E}o}{wL}\cos wt$

$$= -\frac{\mathcal{E}o}{wL}\sin(\pi/2 - wt)$$

$$I = I_o \sin(wt - \pi/2)$$

14. Capacitive reactance is the opposition offered by the inductor in flow of ac. It is denoted by $X_C$.

Let $\mathcal{E} = \mathcal{E}_o \sin w\, t$ be the emf of ac source connected with inductor in the circuit

Due to the continuous charging and discharging of capacitor plates, a continuous but alternating current exist in the circuit.

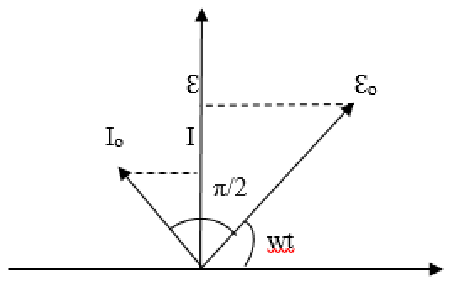
At any time t, Potential difference across the capacitor plates = applied emf

$V = \mathcal{E} = \mathcal{E}_o \sin w\, t$

But $V = Q/C$

Or $Q = CV = C\, \mathcal{E}_o \sin w\, t$

Current at instant t will be,



$I = \frac{dQ}{dt} = d\,(C\,\mathcal{E}_o \sin w\,t)/dt$

$I = w\,C\,\mathcal{E}_o \cos w\,t$

$I = I_o \cos wt$

$I = I_o \sin (wt + \pi/2)$

15. R remains unaffected, inductive reactance get tripled and capacitive reactance become one third of the original value

16.

| RESISTANCE | REACTANCE | IMPEDANCE |
|---|---|---|
| It can be seen in both ac and dc circuits | It can be seen only in ac circuits | It can be seen only in ac circuits |
| Happen due to resistor in a circuit | Happen due to inductor or capacitor in a circuit | Happen due to resistor and inductor or capacitor or both in a circuit |
| Represented by R | Represented by X | Represented by Z |
| It doesn't have a phase angle | It has a phase angle | It has a phase angle |

17. (a)

| INDUCTIVE REACTANCE | CAPACITIVE REACTANCE |
|---|---|
| Opposition offered by the inductor in flow of current | Opposition offered by the capacitor in flow of current |
| Depends directly on frequency of ac | Depends inversely on frequency of ac. |
| Allow dc | Block dc |

(b) $Z^2 = R^2 + X_L^2$, if frequency of applied signal is decreased, the term $X_L = \omega L$ also decreased and as a result value of impedance decreased consequently.

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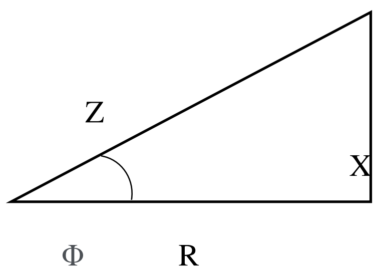
# SECTION B
# COMPETENCY BASED QUESTIONS

Q1. Quantity that measures the opposition offered by a circuit to the flow of current is called impedance. For an ac circuit, reactance corresponding to inductor and capacitor is also considered, along with resistance. So, impedance for the ac circuit can be given mathematically as

$Z^2 = R^2 + X^2$, where X is the reactance.

Impedance triangle is a right-angled triangle




given as in the diagram and satisfy the above mathematical equation.

Also phase angle $\Phi$ between current and voltage can be given by using this impedance triangle

(i) Impedance for a purely capacitive circuit depends on

(a) f (b) 2f (c) 1/f (d) 1/2f

(ii) Impedance for a purely inductive circuit depends on

(a) f (b) 2f (c) 1/f (d) 1/2f

(iii) For ac circuit containing resistor only, value of $\Phi$ will be

(a)0 (b) $\pi/2$ (c) - $\pi/2$ (d) $\pi$

(iv) What is the impedance of a capacitor with capacitance C in an ac circuit having source of frequency 50 Hz

(a) 1/C (b) 1/50 C (c)1/100 C (d) 1/314 C

(v) What are the dimensions of impedance?

## Assertion & Reasoning

(a) Assertion A and reason R are both true and R is the correct explanation of A

(c) Assertion A and reason R are both true but R is not the correct explanation of A

(d) Assertion A is true but reason R is false

(e) Both assertion A and reason R are false

Q2.A: Current in purely inductive or purely capacitive circuit is wattless.

R: No power is dissipated when current flows through a purely inductive or capacitive circuit

Q3.A: If a capacitor is connected to a dc source, it will have infinite reactance.

R: Reactance of capacitor is directly proportional to frequency.

Q4.A: At resonance, impedance is minimum.

R: For an ac circuit, inductive and capacitive reactance are equal and opposite at resonance.

Q5.For an inductive circuit with zero resistance, the current lags behind the applied voltage by an angle……….

$0^o$ b) $30^o$ c)$60^o$ d) $90^o$

## SA-I (2 MARKS EACH)

Q6. Which one is more dangerous----- a 220 V ac or a 220 V dc, explain?

Q7. Why capacitor block dc but pass ac through it, explain?

Q8. Why inductor provides an easy way to dc while resistive to ac?

## SA-II (3 MARKS EACH)

Q9. A 50 μF capacitor is connected to a 100V, 50 Hz ac supply

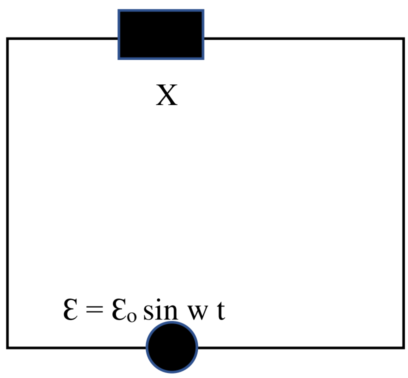
(i)Determine rms value of current

(ii)What is the net power absorbed by circuit in one complete cycle.

## LA (5 MARKS)

Q10. If X, an unknown circuit element is connected to the source as shown in the diagram, such that the current through X is given as

$I = I_o \sin(wt + \pi/2)$




(i)Identify the device X?

(ii)Write expression for the reactance of X.

(iii) Draw phasor diagram for device X

(iv) Draw a graph showing variation of reactance of device X with frequency.

(v) What happen if the source is replaced by a dc source?

## SECTION B-ANSWER KEY

1.(i) c) 1/f

(ii) a) f

(iii) a) 0

(iv) d) 1/314 C

(V) $M^1 L^2 T^{-3} A^{-2}$

2.a) Assertion A and reason R are both true and R is the correct explanation of A.

3.c) Assertion A is true but reason R is false

4.a) Assertion A and reason R are both true and R is the correct explanation of A.

5.d) $90^o$

6.peak value of 220 V ac is 220 $\sqrt{2}$ V = 311 V while for 220 V dc it is 220 V so, 220 V ac is more dangerous.

7. For a capacitor, capacitive reactance is $X_c = 1/2\pi f C$ means capacitive reactance depends inversely on frequency f

8.For dc with zero frequency, capacitor show infinite reactance and block dc while allow ac to pass through it.

For an inductor, inductive reactance is $X_L = 2\pi f L$

So, for dc with zero frequency, inductor will show zero reactance and give it easy way to pass.

While for ac with some finite frequency, inductive reactance will show some finite value and cause opposition in the flow of current.

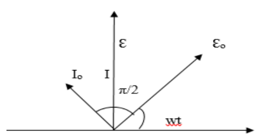
9.C = 50 uF, $V_{rms}$ = 100 V, f = 50 Hz

$I_{rms} = V_{rms} / X_c = 2\pi f C V_{rms} = 1.57$ A

$P_{av} = V_{rms} . I_{rms} . \cos \pi/2 = 0$

10. (i) Capacitor (ii) $X_C = 1/wC$

(iii)



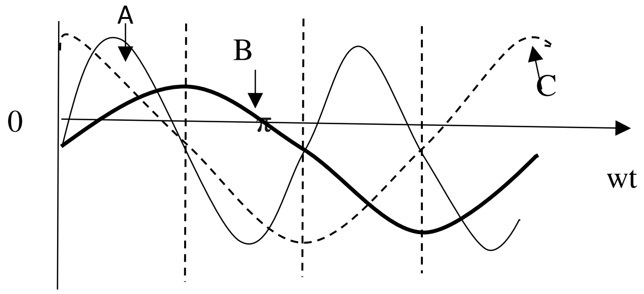

(iv)

(v) for dc, f = 0 so w = 0, hence capacitor block dc

# SECTION C

## COMPETENCY BASED QUESTIONS

Q1. A device X is connected to an alternating source $\mathcal{E} = \mathcal{E}_0 \sin w\, t$. The variation of voltage, current and power in one cycle is shown in the graph




i) Identify X

ii) Which of the curves represent voltage, current and power consumed in the circuit

iii) How does its impedance vary with frequency?

## SA-I (2 MARKS EACH)

Q2.“An alternating current of frequency 15 cps can be used for lighting purpose”, true/ false. Justify your answer.

Q3.“For a very high frequency, capacitor behaves as a conductor”, true/ false. Justify your answer.

Q4.“At resonance, inductive reactance is equal to capacitive reactance”, true/ false. Justify your answer.

Q5.“An alternating current doesn’t show any magnetic effect”, true/ false. Justify your answer.

## SA-II (3 MARKS EACH)

Q6.An alternating source of 220 V is connected to a circuit having a device “A”, a current of 0.5 A flows, which lag behind the applied voltage in phase by $\pi/2$. If the same voltage is applied to another device “B”, same current flows but now it is in phase with the applied voltage.

(i) Name the devices A and B

(ii) Draw the phasor diagram for device A

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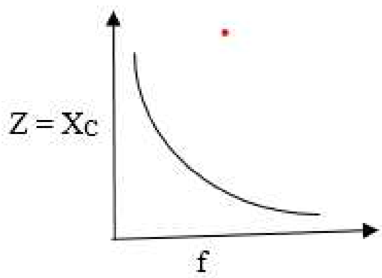
## SECTION C-ANSWERS

1.(i) Device X is a capacitor.

(ii) curve A represents power; curve B represents voltage and curve C represents Current

(iii)




2.true, in this case fluctuation will be so rapid that due to persistence of vision bulb seems glowing

3.true, at very high frequency, capacitive reactance become negligibly small and capacitor behaves like a pure conductor.

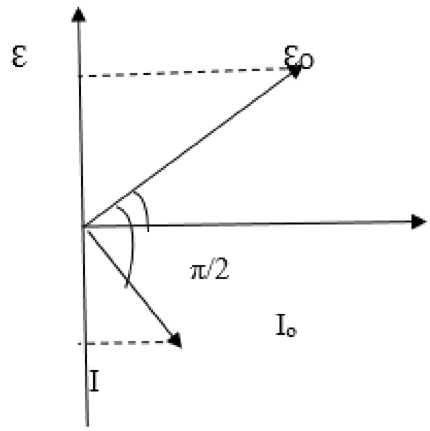
4.True, at resonance value of current is maximum and impedance is minimum, which is possible if inductive reactance and capacitive reactance are equal.

5.False, an alternating current produces a magnetic field whose magnitude and direction changes periodically.

6.i) A is inductor

B is resistor

ii) Phasor diagram

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## QUESTIONS ONLY

**Question Types Includes: 1) MCQ 2) VSA 3) SA-I 4) SA-II 5) LA 6) Assertion and reason 7) Case study.**

# SECTION A

### MULTIPLE CHOICE QUESTION

Q1.Power delivered by the source of the circuit becomes maximum, when

(a) $\omega L = \omega C$ (b) $\omega L = 1/\omega C$ (c) $\omega L = -(1/\omega C)^2$ (d) $\omega L = \sqrt{\omega C}$

Q2. The power factor of LCR circuit at resonance is

(a) 0.707 (b) 1 (c) Zero (d) 0.5

Q3. The phase difference between the current and voltage of LCR circuit in series combination at resonance is

(a) 0 (b) $\pi / 2$ (c) $\pi$ (d) $-\pi$

Q4. What will be the phase difference between virtual voltage and virtual current, when the current in the circuit is wattless

(a) 90° (b) 45° (c) 180° (d) 60°

**Assertion Reasoning Question:**

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false

Q5. **Assertion** : A bulb connected in series with a solenoid is connected to ac source. If a soft iron core is introduced in the solenoid, the bulb will glow brighter.

**Reason** : On introducing soft iron core in the solenoid, the inductance decreases.

Q6. Which of the following is constructed on the principle of electromagnetic induction:

(a) Galvanometer (b) a.c. generator (c) Generator (d) Voltmeter

Q7. A transformer is based on the principle of

(a) Mutual inductance (b) Self-inductance (c) Ampere's law (d) Lenz's law

Q8. A transformer is employed to

(a) Obtain a suitable dc voltage (b) Convert dc into ac

(c) Obtain a suitable ac voltage (d) Convert ac into dc

Q9. Quantity that remains unchanged in a transformer is

(a) Voltage (b) Current (c) Frequency (d) None of above

Q10. **Assertion**: Soft iron is used as a core of transformer.

**Reason**: Area of hysteresis is loop for soft iron is small

## SHORT ANSWER TYPE I (2MARKS EACH)

Q11.What is wattless current?

Q12.Mention the two characteristic properties of the material suitable for making core of a transformer.

Q13.A generator developed an emf of 120V and has terminal potential difference of 115V, when the armature current is 25A. What is the resistance of armature?

## SHORT ANSWER TYPE II (3MARKS EACH)

Q14.A circuit is set up by connecting inductance L = 100 mH, resistor R = 100 Ω and a capacitor of reactance 200 Ω in series. An alternating emf of $150\sqrt{2}$ V, 500/π Hz is applied across this series combination. Calculate the power dissipated in the resistor.

## LONG ANSWER TYPE (5MARKS EACH)

Q15. (a) What is impedance?

(b) A series LCR circuit is connected to an ac source having voltage $V = V_0 \sin \omega t$ . Derive expression for the impedance, instantaneous current and its phase relationship to the applied voltage. Find the expression for resonant frequency.

Q16. Explain with the help of a labelled diagram, the principle and working of an ac generator. Write the expression for the emf generated in the coil in terms of speed of rotation. Can the current produced by an ac generator be measured with a moving coil galvanometer?

Q17. Describe briefly, with the help of a labelled diagram, the basic elements of an ac generator. State its underlying principle. Show diagrammatically how an alternating emf is generated by a loop of wire rotating in a magnetic field. Write the expression for the instantaneous value of the emf induced in the rotating loop.

Q18. State the working of ac generator with the help of a labelled diagram. The coil of an ac generator having N turns, each of area A, is rotated with a constant angular velocity ω. Deduce the expression for the alternating emf generated in the coil. What is the source of energy generation in this device?

Q19. (a) Describe briefly, with the help of a labelled diagram, the working of a step-up transformer.

(b) Write any two sources of energy loss in a transformer.

(c) A step-up transformer converts a low voltage into high voltage. Does it not violate the principle of conservation of energy? Explain.

## CASE STUDY TYPE (4 MARKS EACH)

Q20. When electric power is transmitted over great distances, it is economical to use a high voltage and a low current to minimize the $I^2R$ loss in the transmission lines. Consequently, 350-kV lines are common, and in many areas even higher-voltage (765-kV) lines are under construction. At the receiving end of such lines, the consumer requires power at a low voltage (for safety and for efficiency in design). Therefore, a device is required that can change the alternating voltage and current without causing appreciable changes in the power delivered. The ac transformer is that device. In its simplest form, the ac transformer consists of two coils of wire wound around a core of iron. The coil on the left, which is connected to the input alternating voltage source and has $N_1$ turns, is called the primary winding (or the primary). The coil on the right, consisting of $N_2$ turns and connected to a load resistor R, is called the secondary winding (or the secondary). The purpose of the iron core is to increase the magnetic flux through the coil and to provide a medium in which nearly all the flux through one coil passes through the other coil. Eddy current losses are reduced by using a laminated core. Iron is used as the core material because it is a soft ferromagnetic substance and hence reduces hysteresis losses. Typical transformers have power efficiencies from 90% to 99%. In the discussion that follows, we assume an ideal transformer, one in which the energy losses in the windings and core are zero.

1. Name the different types of losses involve in transformer.
2. How to minimise eddy current losses in transformer?
3. What cause the Hysteresis loss?
4. Which type of transformer used at the receiving end?

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# ANSWERS-SECTION A

**Question Types Includes 1) MCQ 2) VSA 3) SA-I 4) SA-II 5) LA 6) Assertion and reason 7) Case study.**

## MULTIPLE CHOICE QUESTION

1. (a) ωL =ωC
2. (b) 1
3. (a) 0
4. (a) 90°
5. (d) If the assertion and reason both are false
6. (b) a.c. generator
7. (a) Mutual inductance
8. (c) Obtain a suitable ac voltage
9. (c) Frequency
10. (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

## SHORT ANSWER TYPE I (2MARKS EACH)

11. When pure inductor and/or pure capacitor is connected to ac source, the current flows in the circuit, but with no power loss; the phase difference between voltage and current is $\pi/2$.
12. Two characteristic properties: (i) Low hysteresis loss (ii) Low coercivity
13. R= E-V/I = 0.2Ω.

## SHORT ANSWER TYPE II (3MARKS EACH)

14. $X_L = \omega L = 2\pi\nu L = 100\Omega$

$Z = \sqrt{R^2 + (Xc - XL)^2} \quad = 100\sqrt{2}\ \Omega$

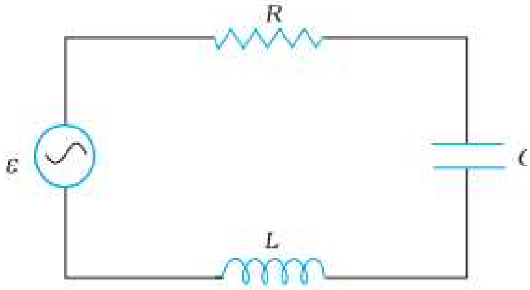
$P_{dissipated} = (Vrms/Z)^2 R = 225W$

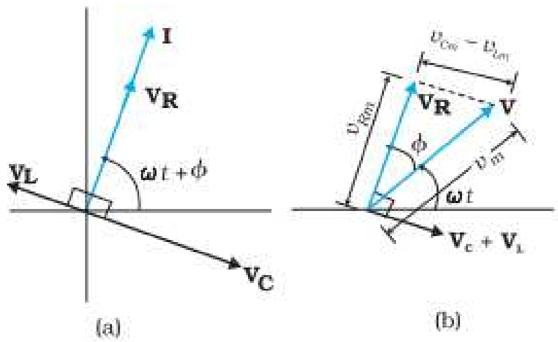
## LONG ANSWER TYPE (5MARKS EACH)

15.(a) The opposition offered by the combination of a resistor and reactive component to the flow of ac is called impedance. Mathematically it is the ratio of rms voltage applied and rms current produced in circuit i.e., $Z= \frac{Vrms}{\mathrm{Irms}}$ . Its unit is ohm (Ω)

(b)

From the circuit shown in Fig. , the resistor, inductor and capacitor are in series. Therefore, the ac current in each element is the same at any time, having the same amplitude and phase. Let it be i = $i_m \sin(\omega t+\varphi)$ where $\varphi$ is the phase difference between the voltage across the source and the current in the circuit




Let **I** be the phasor representing the current in the circuit. Further, let $\mathbf{V_L}$**,** $\mathbf{V_R}$**,** $\mathbf{V_C}$**,** and **V** represent the voltage across the inductor, resistor, capacitor and the source, respectively. We know that $\mathbf{V_R}$ is parallel to **I**, $\mathbf{V_C}$ is $\pi/2$ behind **I** and $\mathbf{V_L}$ is $\pi/2$ ahead of **I**. $\mathbf{V_L}$**,** $\mathbf{V_R}$**,** $\mathbf{V_C}$**,** and **I** are shown in Fig. with appropriate phase-relations.

The length of these phasors or the amplitude of $\mathbf{V_L}$**,** $\mathbf{V_R}$**,** and $\mathbf{V_C}$**,** are:

$V_{Rm} = i_m R$, $v_{Cm} = i_m X_C$, $v_{Lm} = i_m X_L$

From phasor diagram we have

$\mathbf{V_L + V_R + V_C = V}$

Since $\mathbf{V_C}$ and $\mathbf{V_L}$ are always along the same line and in opposite directions, they can be combined into a single phasor **(**$\mathbf{V_C + V_L}$) which has a magnitude $|V_{cm} - V_{Lm}|$Since **V** is represented as the hypotenuse of a right-traingle whose sides are $\mathbf{V_R}$ and ($\mathbf{V_C + V_L}$),

The Pythagorean Theorem gives

$$V_m^2 = V_{Rm}^2 + (V_{Cm} - V_{Lm})2$$

$$V_m^2 = (i_m R)^2 + (i_m X_C - i_m X_L)2$$

$$V_m^2 = i_m^2[(R)^2 + (X_C - X_L)2]$$

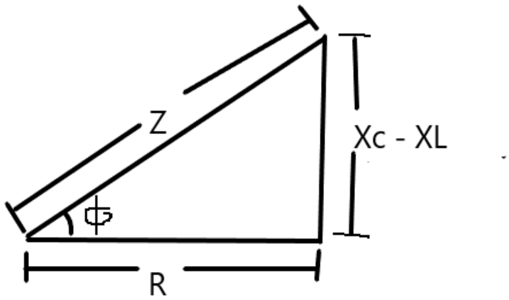
$i_m = V_m / \sqrt{R^2 + (Xc - X_L)^2}$

$i_m = V_m / Z$

$Z=\sqrt{R^2 + (Xc - X_L)^2}$

Where Z is called impedance of the LCR series circuit which is opposition offered to the current in LCR series circuit.

And $\tan \varphi = \frac{V_{Cm} - V_{Lm}}{V_{Rm}} = \frac{X_C - X_L}{R}$




Impedance Triangle

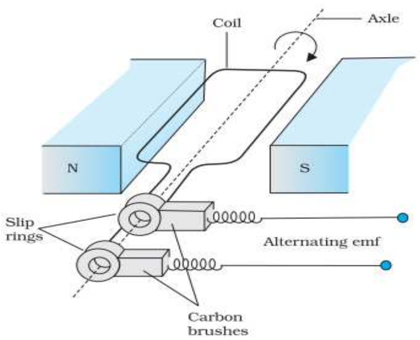
**Instantaneous Current** $I = \frac{V_m \sin wt + \varphi}{\sqrt{R^2 + (Xc - X_L)^2}}$

**At resonance** $X_C = X_L$

$\frac{1}{\omega C} = \omega L$

**Resonance frequency** $\omega_r = \frac{1}{\sqrt{LC}}$

16. Principle: Rotating coil kept in magnetic field produce ac current.




**Construction**: It consists of the four main parts:

**Field Magnet:** It produces the magnetic field.

**Armature:** It consists of a large number of turns of insulated wire in the soft iron drum or ring. It can revolve round an axle between the two poles of the field magnet. The drum or ring serves the two purposes: (a) It serves as a support to coils and (b) It increases the magnetic field due to air core being replaced by an iron core.

**Slip Rings**: The slip rings are the two metal rings to which the ends of armature coil are connected. These rings are fixed to the shaft which rotates the armature coil so that the rings also rotate along with the armature.

**Brushes**: These are two flexible metal plates or carbon rods) which are fixed and constantly touch the revolving rings. The output current in external load $R_L$ is taken through these brushes.

**Expression for induced emf**

When the coil is rotated with a constant angular speed ω, the angle θ between the magnetic field vector B and the area vector A of the coil at any instant t is θ = ωt (assuming θ = 0° at t = 0). As a result, the effective area of the coil exposed to the magnetic field lines changes with time, the flux at any time t is $\Phi_B$ = BA cos θ = BA cos ωt

From Faraday's law, the induced emf for the rotating coil of N turns is then,

$$E = -N\frac{d\phi_B}{dt} = -NBA\frac{d\,cos\omega t}{dt}$$

Thus, the instantaneous value of the emf is

ε = NBA ω sin ωt

where NBAω is the maximum value of the emf, which occurs when sin ωt = ±1.

If we denote NBAω as $\varepsilon_0$,

then $\varepsilon = \varepsilon_0 \sin \omega t$

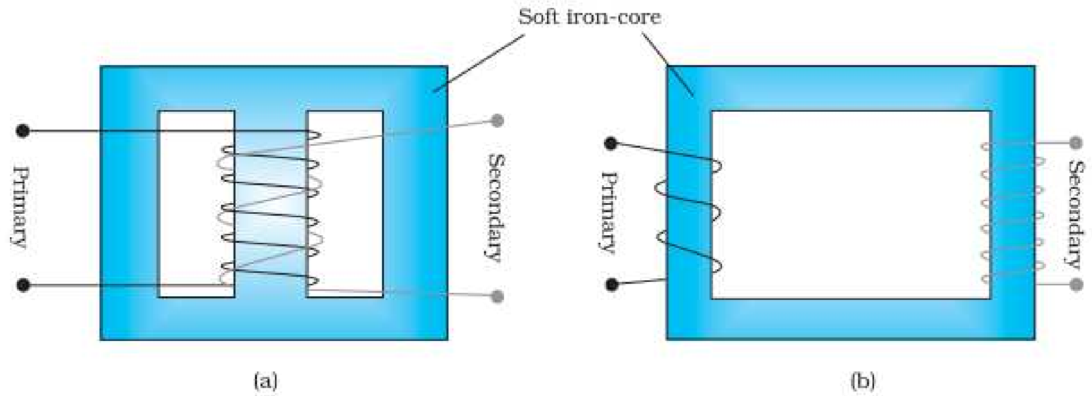
Obviously, the emf produced is alternating and hence the current is also alternating. Current produced by an ac generator cannot be measured by moving coil ammeter; because the average value of ac over full cycle is zero

17. Same as question no 2

18. Same as question no 2

The source of energy generation is the mechanical energy of rotation of armature coil.

19.




**Principle** : Based on Mutual inductance.

**Construction** : A transformer consists of two sets of coils, insulated from each other. They are wound on a soft-iron core, either one on top of the other or on separate limbs of the core. One of the coils called the primary coil has N p turns. The other coil is called the secondary coil; it has Ns turns. Often the primary coil is the input coil and the secondary coil is the output coil of the transformer

**Working** : When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it. The value of this emf depends on the number of turns in the secondary. We consider an ideal transformer in which the primary has negligible resistance and all the flux in the core links both primary and secondary windings.

Let φ be the flux in each turn in the core at time t due to current in the primary when a voltage $v_p$ is applied to it.

Induced emf $E_S$ or voltage in secondary coil of Ns turns will be

$$\varepsilon_S = -N_s \frac{d\phi}{dt}$$

The alternate flux φ induced back emf in the primary coil is

$$\varepsilon_p = -N_p \frac{d\phi}{dt}$$

But $\varepsilon_p = v_p$. If this were not so, the primary current would be infinite since the primary has zero resistance (as assumed). If the secondary is an open circuit or the current taken from it is small, then to a good approximation $\varepsilon_s = v_s$

$v_{s=-\ N_s}\ \frac{d\phi}{dt}$ and $v_{p=-\ N_p}\ \frac{d\phi}{dt}$

Thus

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

If the transformer is assumed to be 100% efficient (no energy losses), the power input is equal to the power output, and since p = i v,

$$V_{\ p}\, i_p\ = V_s i_s$$

Thus $\frac{i_p}{i_c}\ = \frac{V_s}{V_p}\ = \frac{N_s}{N_p}$

**(b)Transformer Losses**

Flux Leakage: There is always some flux leakage; that is, not all of the flux due to primary passes through the secondary due to poor design of the core or the air gaps in the core. It can be reduced by winding the primary and secondary coils one over the other.

Resistance of the windings: The wire used for the windings has some resistance and so, energy is lost due to heat produced in the wire ($I^{\ 2}$ R). In high current, low voltage windings, these are minimised by using thick wire.

Eddy currents: The alternating magnetic flux induces eddy currents in the iron core and causes heating. The effect is reduced by having a laminated core.

Hysteresis: The magnetisation of the core is repeatedly reversed by the alternating magnetic field. The resulting expenditure of energy in the core appears as heat and is kept to a minimum by using a magnetic material which has a low hysteresis loss.

(b) When output voltage increases, the output current automatically decreases to keep the power same. Thus, there is no violation of conservation of energy in a step up / step down transformer.

## CASE STUDY TYPE (4 MARKS EACH)

20. Ans 1. Copper loss, Eddy current loss, Hysteresis loss, flux loss

Ans2. Eddy current losses are reduced by using a laminated core.

Ans3. Choosing a material having small Hysteresis loop area

Ans 4. Step down transformer.

********

## SECTION B

## MULTIPLE CHOICE QUESTION

Q1. In LCR circuit, the capacitance is changed from C to 4C. For the same resonant frequency, the inductance should be changed from L to

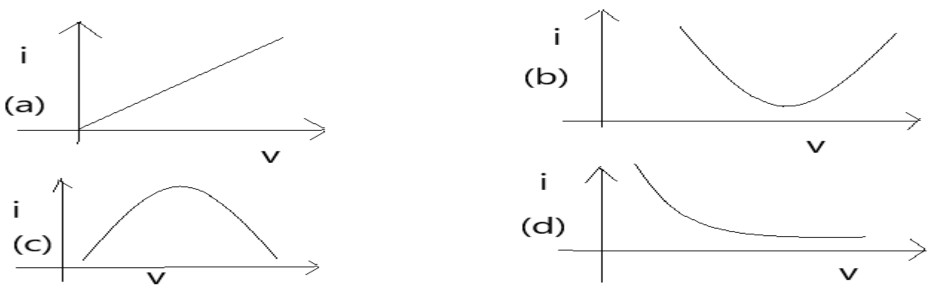
(a) 2L (b) L / 2 (c) L / 4 (d) 4 L

Q2. A bulb and a capacitor are connected in series to a source of alternating current. If its frequency is increased, while keeping the voltage of the source constant, then

(a) Bulb will give more intense light (b) Bulb will give less intense light

(c) Bulb will give light of same intensity as before (d) Bulb will stop radiating light

Q3. The i - ν curve for anti-resonant circuit is




**Assertion-Reasoning Question:**

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false

Q4. **Assertion**: When capacitive reactance is smaller than the inductive reactance in LCR current, e.m.f. leads the current.

**Reason**: The phase angle is the angle between the alternating e.m.f and alternating current of the circuit.

Q5. **Assertion**: A capacitor of suitable capacitance can be used in an ac circuit in place of the choke coil.

**Reason**: A capacitor blocks dc and allows ac only.

Q6. If rotational velocity of an a.c. generator armature is doubled, then induced e.m.f will become

(a) Half (b) Two times (c) Four times (d) Unchanged

Q7. In a step-up transformer, the turn ratio is 1: 2. A Leclanche cell (e.m.f. 1.5V) is connected across the primary. The voltage developed in the secondary would be

(a) 3.0 V (b) 0.75 V (c) 1.5 V (d) Zero

Q8. In a step-up transformer the turn ratio is 1:10. A resistance of 200 ohm connected across the secondary is drawing a current of 0.5 A. What is the primary voltage and current?

(a) 50 V, 1 amp (b) 10 V, 5 amp (c) 25 V, 4 amp (d) 20 V, 2 amp

Q9. **Assertion**: A given transformer can be used to step-up or step-down the voltage.

**Reason**: The output voltage depends upon the ratio of the number of turns of the two coils of the transformer.

## SHORT ANSWER TYPE I (2MARKS EACH)

Q10.Define power factor. State the conditions under which it is (i) maximum and (ii) minimum.

Q11.In a series LCR circuit with an ac source of effective voltage 50 V, frequency $\nu = 50/\pi$ Hz, R = 300 Ω, C = 20 μF and L = 1.0 H. Find the rms current in the circuit.

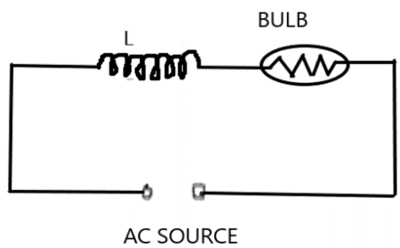
Q12.An output voltage of an ideal transformer connected to 240 V ac mains is 24V. When the transformer used to light a bulb 24V, 24W Calculate the current in the primary of the circuit.

## SHORT ANSWER TYPE II (3MARKS EACH)

Q13.A series LCR circuit is connected to an ac source (200 V, 50 Hz). The voltages across the resistor, capacitor and inductor are respectively 200 V, 250 V and 250 V.

(i)The algebraic sum of the voltages across the three elements is greater than the voltage of the source. How is this paradox resolved?

(ii)Given the value of the resistance of R is 40 W, calculate the current in the circuit.




Q14.An inductor L of reactance $X_L$ is connected in series with a bulb B to an ac source as shown in figure. Explain briefly how does the brightness of the bulb change when (i) number of turns of the inductor is reduced (ii) an iron rod is inserted in the inductor and (iii) a capacitor of reactance $X_C = X_L$ is included in the circuit.

## LONG ANSWER TYPE (5MARKS EACH)

Q15. (a) An ac source of voltage $V = V_0 \sin \omega t$ is connected to a series combination of L, C and R. Use the phasor diagram to obtain expressions for impedance of the circuit and phase angle between voltage and current. Find the condition when current will be in phase with the voltage. What is the circuit in this condition called?

(b) In a series L R circuit $X_L = R$ and power factor of the circuit is $P_1$. When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes $P_2$. Calculate $P_1 / P_2$ .

Q16. (a) An alternating voltage $V = V_m \sin \omega$ applied to a series LCR circuit drives a current given by $i = i_m \sin (\omega t + \varphi)$ . Deduce an expression for the average power dissipated over a cycle.

(b) For circuits used for transporting electric power, a low power factor implies large power loss in transmission. Explain.

Q17. Draw a schematic diagram of a step-up transformer. Explain its working principle. Deduce the expression for the secondary to primary voltage in terms of the number of turns in the two coils. In an ideal transformer, how is this ratio related to the currents in the two coils? How is the transformer used in large scale transmission and distribution of electrical energy over long distances?

## CASE STUDY TYPE (4 MARKS EACH)

Q18. An airport metal is essentially a resonant circuit. The portal you step through is an inductor (a large loop of conducting wire) that is part of the circuit. The frequency of the circuit is tuned to the resonant frequency of the circuit when there is no metal in the inductor. Any metal on your body increases the effective inductance of the loop and changes the current in it. When you pass through a metal detector, you become part of a resonant circuit. As you step through the detector, the inductance of the circuit changes, and thus the current in the circuit changes.

1. What is resonance?
2. On what factors does resonance frequency depends?
3. For the metal detector to detect a small metal object the sharpness of the current versus frequency graph be more or less? Justify your answer.
4. What is impedance of the circuit at resonance?

**********

## ANSWERS-SECTION B

**Question Types Includes 1) MCQ 2) VSA 3) SA-I 4) SA-II 5) LA 6) Assertion and reason 7) Case study.**

## MULTIPLE CHOICE QUESTION

1. Ans. (c) L / 4

2. Ans. (a) Bulb will give more intense light

3. Ans. (b)

4. Ans. (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

5.Ans. (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

6. Ans. (b) Two times

7. Ans. (d) Zero

8. Ans. (b) 10 V, 5 amp

9. Ans. (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

## SHORT ANSWER TYPE I (2MARKS EACH)

10. The power factor (cos φ) is the ratio of resistance and impedance of an ac circuit i.e., Power factor, cos φ= R/ Z Maximum power factor is 1 when Z = R i.e., when circuit is purely resistive. Minimum power factor is 0 when R = 0 i.e., when circuit is purely inductive or capacitive

11. Irms = Vrms/ Z

Here Z=$\sqrt{R^2 + (Xc - XL)^2}$ $X_L$= ωL = 100Ω and Xc= 1/ ωC = 500Ω

Z= 500Ω Irms = 0.1A

12. $I_2$=$P_2$/$E_2$ = 1A

## SHORT ANSWER TYPE II (3MARKS EACH)

13. Ans.(i) $V_{eff} = V_R + V_L + V_C$ = 200 V + 250 V + 250 V = 700V which is greater than 200V . This is due to $V_R$,$V_L$ and $V_C$ are vectors and cannot add like a numbers

Thus Veff = $\sqrt{{V_R}^2 + (V_L - V_C)^2}$ Veff = $\sqrt{(200)^2 + (250 - 250)^2}$

Veff= 200V.

14. (i) When the number of turns in the inductor is reduced, the self-inductance of the coil decreases; so impedance of circuit reduces and so current in the circuit increases. Thus, the brightness of the bulb increases. (ii) When iron (being a ferromagnetic substance) rod is inserted in the coil, its inductance increases and in turn, impedance of the circuit increases. As a result, a larger fraction of the applied ac voltage appears across the inductor, leaving less voltage across the bulb. Hence, brightness of the bulb decreases. (iii) When capacitor of reactance $X_C = X_L$ is introduced, the net reactance of circuit becomes zero, so impedance of circuit decreases; it becomes Z = R; so current in circuit increases; hence brightness of bulb increases. Thus brightness of bulb in both cases increases

## LONG ANSWER TYPE (5MARKS EACH)

15. (a) Refer Answer Section A1

(b) Case1. When $X_L = R$ then $Z= \sqrt{2}R$ thus power factor $P1 = \frac{R}{Z}$ becomes $P1 = 1/\sqrt{2}$

Case 2. When $X_L = X_C$ then power factor $P_2 = 1$

Thus $P_1/ P_2 = 1/\sqrt{2}$

16. In LCR series circuit

$V= Vm \sin\omega t$ and $i = i_m \sin(\omega t+\varphi)$

Thus instantaneous power $P= Vi = (Vm\sin\omega t)(i_m \sin(\omega t+\varphi))$

$= (V_m i_m \sin\omega t)[\sin\omega t \cos\varphi + \cos\omega t \sin\varphi] = V_m i_m [\sin^2\omega t \cos\varphi + \cos^2 \omega t \sin\varphi]$

Thus Average power over a complete cycle

$Pav = \frac{1}{T}\int_0^T Pdt = \frac{1}{T}\int_0^T \text{Vmim} [\text{Sin}2\omega t \cos\varphi + \cos 2\,\omega t \sin\varphi]dt$

Solving

$\int_0^T sin^2\omega t cos\phi dt$ = T/2 cosφ and $\int_0^T cos^2\omega t sin\phi dt$=0

Thus $Pav = V_m i_m \cos\varphi /2 = V_{rms} I_{rms} \cos\varphi$

(b) $Pav = V_{rms} I_{rms} \cos\varphi$

For small power factor cosφ, $I_{rms}$ increases for constant $V_{rms}$ and power loss is $= I^2_{rms}R$ increases.

17. Derivation is same as answer of Section A5.

The large scale transmission and distribution of electrical energy over long distances is done with the use of transformers. The voltage output of the generator is stepped-up (so that current is reduced and consequently, the $I^2R$ loss is cut down). It is then transmitted over long distances to an area sub-station near the consumers. There the voltage is stepped down. It is further stepped down at distributing sub-stations and utility poles before a power supply of 240 V reaches our homes.

## CASE STUDY TYPE (4 MARKS EACH)

18. Ans1. In LCR series circuit when inductive reactance become equal to capacitive reactance then the current in a circuit become maximum called resonance.

Ans2. Resonance frequency depends on value of inductor L and capacitor C

Ans3. Graph should be sharper for unique resonance frequency.

Ans4. At resonance impedance is minimum i.e. Z= R

****************

## SECTION C

## MULTIPLE CHOICE QUESTION

Q1. An inductance of 1 mH a condenser of 10 μF and a resistance of 50 Ω are connected in series. The reactance of inductor and condensers are same. The reactance of either of them will be

(a) 100 Ω (b) 30 Ω (c) 3.2 Ω (d) 10 Ω

Q2. An ac circuit consists of an inductor of inductance 0.5 H and a capacitor of capacitance 8 μF in series. The current in the circuit is maximum when the angular frequency of ac source is

a) 500 rad/sec (b) $2 \times 10^5$ rad/sec (c) 4000 rad/sec (d) 5000 rad/sec

Q3. An inductive circuit contains a resistance of 10 ohm and an inductance of 2.0 henry. If an ac voltage of 120 volt and frequency of 60 Hz is applied to this circuit, the current in the circuit would be nearly

(a) 0.32 amp (b) 0.16 amp (c) 0.48 amp (d) 0.80 amp

Q4. The power factor of an ac circuit having resistance (R) and inductance (L) connected in series and an angular velocity ω is

(a) $R/\omega L$ (b) $R/(R^2+\omega^2 L^2)^{\frac{1}{2}}$ (c) $\omega L / R$ (d) $R/(R^2 - \omega^2 L^2)^{\frac{1}{2}}$

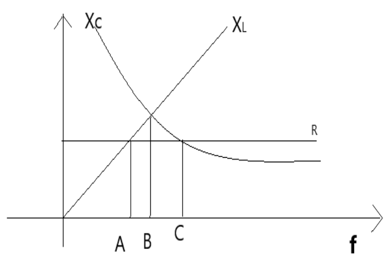
Q5. A telephone wire of length 200 km has a capacitance of 0.014 μF per km. If it carries an ac of frequency 5 kHz, what should be the value of an inductor required to be connected in series so that the impedance of the circuit is minimum

(a) 0.35 mH (b) 35 mH (c) 3.5 mH (d) Zero

Q6. An LCR series circuit with a resistance of 100 ohm is connected to an ac source of 200 V (r.m.s.) and angular frequency 300 rad/s. When only the capacitor is removed, the current lags behind the voltage by 60 °. When only the inductor is removed the current leads the voltage by 60 °. The average power dissipated is

(a) 50 W (b) 100 W (c) 200 W (d)400W

Q7. The figure shows variation of R, $X_L$ and $X_C$ with frequency f in a series L, C, R circuit. Then for what frequency point, the circuit is inductive

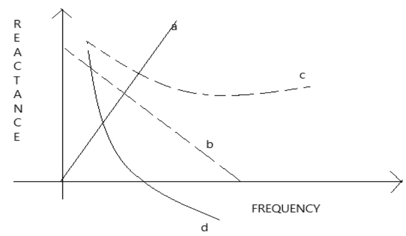



(a) A (b) B (c) C

(d) All points

Q8. Which of the following plots may represent the reactance of a series LC combination

(a) a (b) b (c) c (d) d




Q9. A loss free transformer has 500 turns on its primary winding and 2500 in secondary. The meters of the secondary indicate 200 volts at 8 amperes under these conditions. The voltage and current in the primary is

(a) 100 V, 16 A (b) 40 V, 40 A (c) 160 V, 10 A (d) 80 V, 20 A

Q10. A power transformer is used to step up an alternating e.m.f. of 220 V to 11 kV to transmit 4.4 kW of power. If the primary coil has 1000 turns, what is the current rating of the secondary? Assume 100% efficiency for the transformer

(a) 4 A (b) 0.4 A (c) 0.04 A (d) 0.2 A

## SHORT ANSWER TYPE I (2MARKS EACH)

Q11.An alternating voltage $E = E_0 \sin \omega t$ is applied to a circuit containing a resistor R connected in series with a unknown box. The current in the circuit is found to be $I = I_o \sin (\omega t + \pi/4)$.

(i) State whether the element in the unknown box is a capacitor or inductor.

(ii) Draw the corresponding phasor diagram and find the impedance in terms of R.

Q12.A 60 W load is connected to the secondary of a transformer whose primary draws line voltage. If a current of 0.54 A flows in the load, what is the current in the primary coil? Comment on the type of transformer being used.

Q13. A radio wave of wavelength 300m can be transmitted by a transmission centre. A condenser of capacity 2.4 μF is available. Calculate the inductance of required coil for resonance.

Q14.How much current is drawn by the primary of a transformer which step down 220V to 22V to operate a device with an impedance of 220Ω?

## SHORT ANSWER TYPE II (3MARKS EACH)

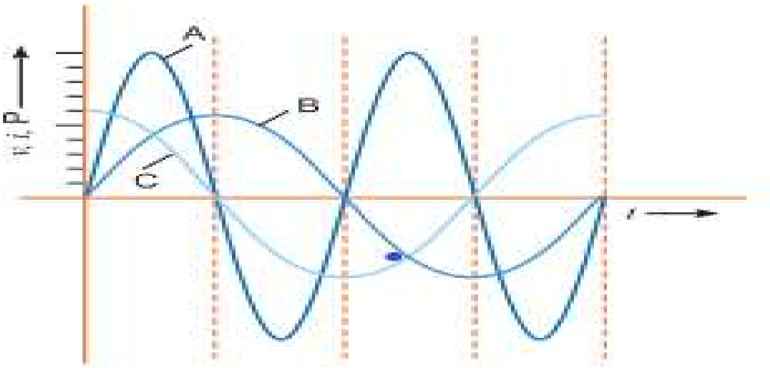
Q15.A capacitor (C) and resistor (R) are connected in series with an ac source of voltage of frequency 50 Hz. The potential difference across C and R is 120 V, 90 V respectively, and the current in the circuit is 3 A. Calculate (i) the impedance of the circuit (ii) the value of the inductance, which when connected in series with C and R will make the power factor of the circuit unity.

Q16.A device ‘X’ is connected to an ac source. The variation of voltage, current and power in one complete cycle is shown in the figure.

(a) Which curve shows power consumption over a full cycle?

(b) What is the average power consumption over a cycle?

(c) Identify the device ‘X’.




## Long Questions (Each carry 5 marks)

Q17. A voltage $V = V_0 \sin \omega t$ is applied to a series LCR circuit. Derive the expression for the average power dissipated over a cycle. Under what condition is (i) no power dissipated even though the current flows through the circuit, (ii) maximum power dissipated in the circuit?

Q18. (a) Draw the diagram of a device which is used to decrease high ac voltage into a low ac voltage and state its working principle. Write four sources of energy loss in this device.

(b) A small town with a demand of 1200 kW of electric power at 220 V is situated 20 km away from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is 0.5 Ω per km. The town gets the power from the line through a 4000-220 V step-down transformer at a sub-station in the town. Estimate the line power loss in the form of heat.

Q19. A 2 μF capacitor, 100 Ω resistor and 8 H inductor are connected in series with an ac source.

(i) What should be the frequency of the source such that current drawn in the circuit is maximum? What is this frequency called?

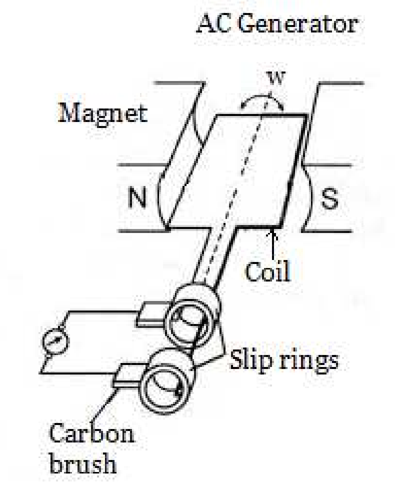
(ii) If the peak value of emf of the source is 200 V, find the maximum current.

(iii)Draw a graph showing variation of amplitude of circuit current with changing frequency of applied voltage in a series LRC circuit for two different values of resistance $R_1$ and $R_2$ ($R_1 > R_2$).

Q20. (i) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary to primary voltage in terms of number of turns and currents in the two coils.
(ii) A power transmission line feeds input power at 2200 V to a step-down transformer with its primary windings having 3000 turns. Find the number of turns in the secondary to get the power output at 220 V.

## CASE STUDY TYPE QUESTIONS (4 MARKS EACH)

Q21. Electric generators are used to produce electrical energy. To understand how they work, let us consider the alternating current (ac) generator, a device that converts mechanical energy to electrical energy. In its simplest form, it consists of a loop of wire rotated by some external means in a magnetic field In commercial power plants, the energy required to rotate the loop can be derived from a variety of sources. For example, in a hydroelectric plant, falling water directed against the blades of a turbine produces the rotary motion; in a coal-fired plant, the energy released by burning coal is used to convert water to steam, and this steam is directed against the turbine blades. As a loop rotates in a magnetic field, the magnetic flux through the area enclosed by the loop changes with time; this induces an emf and a current in the loop according to Faraday's law. The ends of the loop are connected to slip rings that rotate with the loop. Connections from these slip rings, which act as output terminals of the generator, to the external circuit are made by stationary brushes in contact with the slip rings.




1. Name the principle on which ac generator based.
2. What is the condition for maximum induced emf in a Coil?
3. An ac generator consists of 8 turns of wire, each of area A = 0.090 0 $m^2$, and the total resistance of the wire is 12.0 Ω. The loop rotates in a 0.500-T magnetic field at a constant frequency of 60.0 Hz. Find the maximum induced emf.
4. In above case what is the maximum induced current when the output terminals are connected to a low-resistance conductor?

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# ANSWER-KEY

**Question Types Includes 1) MCQ 2) VSA 3) SA-I 4) SA-II 5) LA 6) Assertion and reason 7) Case study.**

## SECTION C

### MULTIPLE CHOICE QUESTION

1. (d) 10 Ω

2. (a) 500 rad/sec

3. (b) 0.16 amp

4. (b) $R/ (R^2 + \omega^2 L^2)^{\frac{1}{2}}$

5. (a) 0.35 mH

6. (d) 400 W

7. (c) C

8. (d) d

9. (b) 40 V, 40 A

10. (b) 0.4 A

## SHORT ANSWER TYPE I (2MARKS EACH)

11(i) As the current leads the voltage by π/4, the element used in black box is a capacitor.

ii) $\tan(\pi/4) = V_C/V_R$ $\quad 1 = V_C/V_R$ $\quad$ gives $X_C = R$

Impedance $Z = \sqrt{R^2 + Xc^2}$ $\quad$ Gives $Z = \sqrt{2}\,R$

12. Here $P_L = 60$ W, $I_L = 0.54$ A . $V_L = 60/0.54 = 111.1$ V

The transformer is step-down and have ½ input voltage.
Hence $I_P = 1/2$ $\quad I_L = 0.27$ A

13. Frequency $\nu = C/\lambda = 10^6$ Hz.

$\upsilon^2 = 1/ 4\pi^2 LC$ $\quad$ Gives $L = 1.055\text{x}10^8$ H

14. $I_2 = E_2/ Z_2 = 0.1$A

$I_1 = E_2\, I_2/E_1 = 0.01$A.
$I_1 = E_2 I_2/E_1 = 0.1$A

## SHORT ANSWER TYPE II (3MARKS EACH)

15. (i)R= $V_R/I$ = 30Ω   Xc = Vc/I = 40Ω  $Z=\sqrt{R^2+Xc^2}$ = 50Ω

(ii) $X_L$ = Xc   ωL= 40  L= 40/ 2πf = 2/5π H.

16.Ans. (a) A

(b) Zero

(c) L or C or LC Series combination of L and C

## Long Questions (Each carry 5 marks)

17. Derivation same as Question B2

(i)Condition for No power loss is Pav = $V_{rms}I_{rms}$ cosφ

Cosφ= 0 i.e. φ= $90^0$ No resistor used in circuit.

(ii)For Maximum power loss $X_C = X_L$ i.e. at resonance. Cosφ =1 and power lost is maximum.

18.(b) Demand of electric power = 1200 kW

Distance of town from power station = 20 km

Two wire = 20 × 2 = 40 km

Total resistance of line = 40 × 0.5 = 20 Ω

The town gets a power of 4000 volts Power = voltage × current

$$I = \frac{1200\text{x}10^4}{4000} = 1200/4 = 300\ A$$

The line power loss in the form of heat = $I^2 \times R$

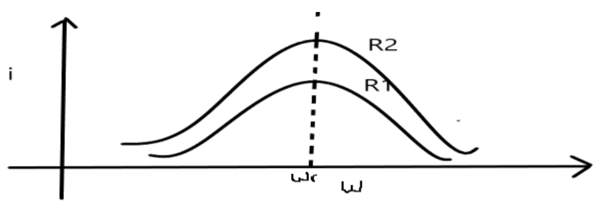
= (300)2 × 20 = 9000 × 20 = 1800 kW

19.

$\omega L = \frac{1}{\omega C}$   $\omega^2 = \frac{1}{LC}$   $2\pi v^2 = \frac{1}{LC}$ $v = 39.80\ Hz$ the frequency is called resonance frequency.

ii.Maximum current at resonance $I_m^{max} = \frac{V_m}{R} = \frac{200}{100} = 2A$

iii.



$R_1$ is greater than $R_2$

20. Ans $N_s = \frac{V_s}{V_p} N_p = 300\ turns$

## CASE STUDY TYPE QUESTIONS (4 MARKS EACH)

21. Answer1. Electromagnetic induction

Answer2. Plane of coil become parallel to magnetic field so $\theta = 90,\ \ sin\theta = 1$

Answer3.$\varepsilon_{max}\ = NBA\ \omega = 136V$

Answer4. $I_{max} = \frac{\varepsilon_{max}}{R}$ = 11.3

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